

30 Days SQL Micro Course

Day 2 – Types of Commands in SQL(DDL)

INSERT Command (DML)

The INSERT command is used to add new records (rows) into a table.

1. INSERT a Single Row

Syntax

```
INSERT INTO table_name (column1, column2, column3)
VALUES (value1, value2, value3);
```

Example

```
INSERT INTO Students (StudentID, Name, Age)
VALUES (1, 'Amit', 20);
```

Result

StudentID	Name	Age
-----------	------	-----

1	Amit	20
---	------	----

2. INSERT Without Column Names

Syntax

```
INSERT INTO table_name
VALUES (value1, value2, value3);
```

Example

```
INSERT INTO Students
VALUES (2, 'Neha', 21);
```

⚠ Values must be in the same order as the table columns.

3. INSERT Multiple Rows

Syntax

```
INSERT INTO table_name (column1, column2, column3)
VALUES
(value1, value2, value3),
(value4, value5, value6);
```

Example

```
INSERT INTO Students (StudentID, Name, Age)
VALUES
(3, 'Rahul', 22),
(4, 'Priya', 19);
```

Result

StudentID	Name	Age
1	Amit	20
2	Neha	21
3	Rahul	22
4	Priya	19

Complete Example

Create Table

```
CREATE TABLE Students (
    StudentID INT,
    Name VARCHAR(50),
    Age INT
);
```

Insert Data

```
INSERT INTO Students
VALUES (1, 'Amit', 20);
```

```
INSERT INTO Students
VALUES (2, 'Neha', 21);
```

Verify

```
SELECT * FROM Students;
```

Output

StudentID	Name	Age
1	Amit	20
2	Neha	21

Common Interview Questions

What is INSERT used for?

Answer: INSERT is used to add new records into a table.

Can we insert multiple rows in one query?

Answer: Yes.

INSERT INTO Students

VALUES

(1, 'Amit', 20),

(2, 'Neha', 21);

Is INSERT a DDL or DML command?

Answer: DML (Data Manipulation Language).

Quick Summary

Command Purpose

INSERT Add new records (rows) to a table

Memory Tip

INSERT = Put/Add new data into a table. ☒

Example:

INSERT INTO Students (StudentID, Name, Age)

VALUES (1, 'Amit', 20);

UPDATE Command (DML)

The **UPDATE** command is used to **modify existing records (rows)** in a table.

1. Update a Single Record

Syntax

```
UPDATE table_name  
SET column_name = value  
WHERE condition;
```

Example

```
UPDATE Students  
SET Age = 21  
WHERE StudentID = 1;
```

Before

StudentID	Name	Age
-----------	------	-----

1	Amit	20
---	------	----

After

StudentID	Name	Age
-----------	------	-----

1	Amit	21
---	------	----

2. Update Multiple Columns

Syntax

```
UPDATE table_name  
SET column1 = value1,  
    column2 = value2  
WHERE condition;
```

Example

```
UPDATE Students  
SET Name = 'Amit Kumar',  
    Age = 22  
WHERE StudentID = 1;
```

Result

StudentID	Name	Age
-----------	------	-----

1	Amit Kumar	22
---	------------	----

3. Update All Records

⚠ Without a WHERE clause, all rows are updated.

Example

```
UPDATE Students  
SET Age = 18;
```

Before

StudentID	Name	Age
-----------	------	-----

1	Amit	20
---	------	----

2	Neha	21
---	------	----

After

StudentID	Name	Age
-----------	------	-----

1	Amit	18
---	------	----

2	Neha	18
---	------	----

Complete Example

Create Table

```
CREATE TABLE Students (  
    StudentID INT,  
    Name VARCHAR(50),  
    Age INT  
);
```

Insert Data

```
INSERT INTO Students  
VALUES (1, 'Amit', 20);
```

```
INSERT INTO Students  
VALUES (2, 'Neha', 21);
```

Update Data

```
UPDATE Students
SET Age = 22
WHERE StudentID = 2;
```

Verify

```
SELECT * FROM Students;
```

Output

StudentID	Name	Age
-----------	------	-----

1	Amit	20
---	------	----

2	Neha	22
---	------	----

Common Interview Questions

What is UPDATE used for?

Answer: UPDATE is used to modify existing records in a table.

Why is WHERE important in UPDATE?

Answer: WHERE specifies which rows should be updated. Without WHERE, all rows will be updated.

Is UPDATE a DDL or DML command?

Answer: DML (Data Manipulation Language).

Quick Summary

Command Purpose

UPDATE	Modify existing records
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Memory Tip

UPDATE = Change existing data in a table. ☒

Example:

```
UPDATE Students
SET Age = 21
WHERE StudentID = 1;
```

DELETE Command (DML)

The **DELETE** command is used to **remove records (rows)** from a table.

1. Delete a Specific Record

Syntax

```
DELETE FROM table_name  
WHERE condition;
```

Example

```
DELETE FROM Students  
WHERE StudentID = 1;
```

Before

StudentID	Name	Age
-----------	------	-----

1	Amit	20
---	------	----

2	Neha	21
---	------	----

After

StudentID	Name	Age
-----------	------	-----

2	Neha	21
---	------	----

Only the row with StudentID = 1 is deleted.

2. Delete Multiple Records

Example

```
DELETE FROM Students  
WHERE Age < 18;
```

This removes all students whose age is less than 18.

3. Delete All Records

⚠ Without a WHERE clause, all rows are deleted.

Example

DELETE FROM Students;

Before

StudentID	Name	Age
-----------	------	-----

1	Amit	20
---	------	----

2	Neha	21
---	------	----

After

StudentID	Name	Age
-----------	------	-----

Result: All rows are removed, but the table still exists.

Complete Example

Create Table

```
CREATE TABLE Students (  
    StudentID INT,  
    Name VARCHAR(50),  
    Age INT  
);
```

Insert Data

```
INSERT INTO Students VALUES (1, 'Amit', 20);  
INSERT INTO Students VALUES (2, 'Neha', 21);  
INSERT INTO Students VALUES (3, 'Rahul', 17);
```

Delete Record

```
DELETE FROM Students  
WHERE StudentID = 3;
```

Verify

```
SELECT * FROM Students;
```

Output

StudentID	Name	Age
-----------	------	-----

1	Amit	20
---	------	----

2	Neha	21
---	------	----

DELETE vs TRUNCATE

Feature	DELETE TRUNCATE	
Removes specific rows	☒	☐
Uses WHERE clause	☒	☐
Removes all rows	☒	☒
Keeps table structure	☒	☒
Faster for all rows	☐	☒

Example

Delete specific rows:

```
DELETE FROM Students  
WHERE Age < 18;
```

Delete all rows quickly:

```
TRUNCATE TABLE Students;
```

Common Interview Questions

What is DELETE used for?

Answer: DELETE removes records from a table.

What happens if WHERE is omitted?

Answer: All rows in the table are deleted.

Is DELETE a DDL or DML command?

Answer: DML (Data Manipulation Language).

Quick Summary

Command Purpose

DELETE Remove records (rows) from a table

Memory Tip

DELETE = Remove selected data from a table. ☒

Example:

```
DELETE FROM Students  
WHERE StudentID = 1;
```

MERGE Command (DML)

The **MERGE** command is used to **INSERT, UPDATE, or DELETE** data based on a **condition**.

It is often called an **UPSERT** operation:

- If a matching record exists → **UPDATE**
- If no matching record exists → **INSERT**

Why Use MERGE?

Instead of writing separate INSERT and UPDATE statements, you can do both in one command.

Syntax

```
MERGE INTO target_table AS T  
USING source_table AS S  
ON condition
```

```
WHEN MATCHED THEN  
    UPDATE SET column1 = value1
```

```
WHEN NOT MATCHED THEN  
    INSERT (column1, column2)  
    VALUES (value1, value2);
```

Example

Target Table: Students

StudentID Name Age

1	Amit	20
2	Neha	21

Source Table: NewStudents

StudentID Name Age

2	Neha	22
3	Rahul	19

MERGE Query

```
MERGE INTO Students AS S
USING NewStudents AS N
ON S.StudentID = N.StudentID
```

```
WHEN MATCHED THEN
  UPDATE SET Age = N.Age
```

```
WHEN NOT MATCHED THEN
  INSERT (StudentID, Name, Age)
  VALUES (N.StudentID, N.Name, N.Age);
```

Result

StudentID Name Age

1	Amit	20
2	Neha	22
3	Rahul	19

What Happened?

- StudentID **2** existed → **Updated**
- StudentID **3** did not exist → **Inserted**

MERGE vs INSERT vs UPDATE

Command Purpose

INSERT	Add new records
UPDATE	Modify existing records
MERGE	Insert or Update based on condition

Common Interview Questions

What is MERGE?

Answer: MERGE combines INSERT and UPDATE operations into a single statement.

Why use MERGE?

Answer: To synchronize data between two tables efficiently.

What is UPSERT?

Answer: Update if the row exists; otherwise insert it. MERGE performs an UPSERT.

Is MERGE a DDL or DML command?

Answer: DML (Data Manipulation Language).

Quick Summary

Command Purpose

MERGE	Insert or Update records based on a condition
-------	---

Memory Tip

MERGE = Match → Update, No Match → Insert ☒

```
MERGE INTO Students
USING NewStudents
ON Students.StudentID = NewStudents.StudentID
WHEN MATCHED THEN UPDATE
WHEN NOT MATCHED THEN INSERT;
```

Interview Questions

INSERT Questions

1. What is the INSERT command?

Answer: INSERT is used to add new records (rows) into a table.

2. What is the syntax of INSERT?

```
INSERT INTO Students (StudentID, Name, Age)
VALUES (1, 'Amit', 20);
```

3. Can you insert multiple rows in one query?

Answer: Yes.

```
INSERT INTO Students
VALUES
(1, 'Amit', 20),
(2, 'Neha', 21);
```

4. What happens if column names are omitted?

Answer: Values must be provided in the exact order of the table columns.

5. Is INSERT a DDL or DML command?

Answer: DML.

UPDATE Questions

6. What is UPDATE used for?

Answer: UPDATE modifies existing records in a table.

7. What is the syntax of UPDATE?

```
UPDATE Students
SET Age = 21
WHERE StudentID = 1;
```

8. Why is WHERE important in UPDATE?

Answer: WHERE specifies which rows to update. Without WHERE, all rows are updated.

9. Can UPDATE modify multiple columns?

Answer: Yes.

```
UPDATE Students
SET Name='Amit Kumar',
    Age=22
WHERE StudentID=1;
```

10. What happens if WHERE is omitted?

Answer: Every row in the table will be updated.

DELETE Questions

11. What is DELETE used for?

Answer: DELETE removes records from a table.

12. What is the syntax of DELETE?

DELETE FROM Students
WHERE StudentID = 1;

13. What happens if WHERE is omitted in DELETE?

Answer: All rows are deleted.

DELETE FROM Students;

14. Does DELETE remove the table structure?

Answer: No. Only data is removed.

15. Is DELETE DDL or DML?

Answer: DML.

MERGE Questions

16. What is MERGE?

Answer: MERGE combines INSERT and UPDATE operations in a single statement.

17. What is UPSERT?

Answer: Update if the row exists; otherwise insert it.

18. Why is MERGE used?

Answer: To synchronize data between two tables efficiently.

19. What happens when records match in MERGE?

Answer: The matched rows are updated.

20. What happens when records do not match?

Answer: New rows are inserted.

Comparison Questions

21. Difference between INSERT and UPDATE?

INSERT	UPDATE
--------	--------

Adds new rows	Modifies existing rows
---------------	------------------------

Creates data	Changes data
--------------	--------------

22. Difference between DELETE and TRUNCATE?

DELETE	TRUNCATE
--------	----------

Can use WHERE	Cannot use WHERE
---------------	------------------

Deletes selected rows	Deletes all rows
-----------------------	------------------

Slower	Faster
--------	--------

DML	DDL
-----	-----

23. Difference between DELETE and DROP?

DELETE	DROP
--------	------

Removes rows	Removes table/database
--------------	------------------------

Structure remains	Structure removed
-------------------	-------------------

DML	DDL
-----	-----

24. Difference between UPDATE and MERGE?

UPDATE	MERGE
--------	-------

Only modifies existing rows	Can update or insert
-----------------------------	----------------------

Works on one table	Usually synchronizes two tables
--------------------	---------------------------------

25. Difference between INSERT and MERGE?

INSERT	MERGE
--------	-------

Only inserts data	Inserts or updates data
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No matching logic	Uses matching condition
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Scenario-Based Questions

26. How would you increase every employee's salary by 10%?

UPDATE Employees
SET Salary = Salary * 1.10;

27. How would you remove students younger than 18?

DELETE FROM Students
WHERE Age < 18;

28. How would you add a new student?

INSERT INTO Students
VALUES (5, 'Priya', 20);

29. How would you update Rahul's age to 21?

UPDATE Students
SET Age = 21
WHERE Name = 'Rahul';

30. How would you synchronize data from NewStudents into Students?

Answer: Use MERGE.

Frequently Asked Viva Questions

31. Which command adds records?

Answer: INSERT

32. Which command modifies records?

Answer: UPDATE

33. Which command removes records?

Answer: DELETE

34. Which command performs UPSERT?

Answer: MERGE

35. Which command is dangerous without WHERE?

Answer: UPDATE and DELETE

36. Can DELETE be rolled back?

Answer: Yes, if the transaction has not been committed.

37. Can TRUNCATE use WHERE?

Answer: No.

38. Which is faster: DELETE or TRUNCATE?

Answer: TRUNCATE.

39. Does DELETE reset identity/auto-increment values?

Answer: Usually no.

40. Does TRUNCATE reset identity values?

Answer: Usually yes (DBMS-dependent).

Top 10 Most Asked Interview Questions

1. What is DML?
2. Difference between INSERT and UPDATE?
3. Difference between DELETE and TRUNCATE?
4. What happens if WHERE is omitted in UPDATE?
5. What happens if WHERE is omitted in DELETE?
6. What is MERGE?
7. What is UPSERT?
8. Can INSERT add multiple rows?
9. Which command modifies existing records?
10. Which command removes records but keeps the table?